

Chapter 2: Prompt Engineering Basics

Class 9 – Prompting for Images & Designs

Learning Objectives

After completing this lesson, students will be able to:

- Understand the basics of AI image generation.
 - Write clear and detailed prompts for AI-generated images.
 - Use descriptive words for objects, people, backgrounds, colors, and lighting.
 - Create prompts for posters, logos, social media graphics, and illustrations.
 - Improve image quality by refining prompts.
 - Apply ethical and responsible practices while creating AI-generated images.
-

Introduction

Artificial Intelligence can create high-quality images, posters, illustrations, logos, and digital artwork from simple text instructions. These text instructions are called **image prompts**.

The quality of an AI-generated image depends on how clearly the prompt is written. A detailed prompt helps AI understand the subject, style, colors, background, lighting, and mood of the image.

Learning how to write effective image prompts helps students create attractive designs for school projects, presentations, competitions, and creative activities.

What is an Image Prompt?

An **Image Prompt** is a written description that tells an AI image generator what kind of picture or design to create.

A good image prompt describes:

- Main subject
- Background
- Colors
- Style

- Lighting
 - Camera angle (optional)
 - Mood
 - Image quality
-

Why are Image Prompts Important?

Good image prompts help to:

- Generate accurate images.
 - Improve creativity.
 - Save design time.
 - Create professional-quality graphics.
 - Produce unique visual content.
 - Improve communication with AI.
-

Components of a Good Image Prompt

A complete image prompt should include:

1. Subject

What should appear in the image?

Example:

"A school student using a laptop."

2. Background

Where is the subject?

Example:

"In a modern classroom."

3. Style

Choose the artistic style.

Examples:

- Realistic
 - Cartoon
 - Digital Art
 - Watercolor
 - 3D Illustration
 - Flat Design
 - Minimalist
-

4. Colors

Mention the preferred color theme.

Examples:

- Blue and White
 - Green Nature Theme
 - Bright Colors
 - Pastel Colors
 - Black and Gold
-

5. Lighting

Examples:

- Natural Light
 - Soft Light
 - Golden Hour
 - Studio Lighting
 - Neon Lighting
-

6. Mood

Examples:

- Happy
 - Professional
 - Creative
 - Inspirational
 - Peaceful
 - Energetic
-

7. Quality

Examples:

- High Resolution
 - 4K
 - Ultra HD
 - Detailed
 - Sharp Focus
-

Writing Effective Image Prompts

Basic Prompt

A tree.

Better Prompt

A large green tree in a beautiful park during sunrise with realistic lighting and high-resolution details.

Professional Prompt

A realistic school campus surrounded by green trees, students walking happily, modern architecture, bright morning sunlight, vibrant colors, ultra-detailed, 4K quality.

Prompting for Posters

Students can use AI to design posters.

Example Prompt

Design a colorful poster for World Environment Day featuring green forests, clean rivers, children planting trees, bold title "Save Nature, Save Future", modern flat illustration style, high resolution.

Prompting for Logos

AI can generate logo concepts.

Example Prompt

Create a modern technology logo with the letter "A", blue and silver color theme, minimal design, professional style, transparent background.

Prompting for Social Media Designs

Students can create banners and social media posts.

Example Prompt

Create an Instagram post promoting Digital Safety with blue background, modern icons, attractive typography, clean layout, professional design.

Prompting for Illustrations

AI can generate illustrations for books, projects, and presentations.

Example Prompt

A colorful illustration of the solar system with labeled planets in a cartoon educational style suitable for school students.

Improving Image Prompts

If the first image is not satisfactory, students can improve the prompt by adding:

- More details.
 - Better colors.
 - Different style.
 - Camera angle.
 - Lighting.
 - Background objects.
 - Higher quality.
-

Example

Before

A village.

After

A peaceful Indian village during sunrise with green fields, traditional houses, farmers working, realistic style, warm lighting, ultra HD.

Ethical Use of AI-generated Images

Students should:

- Respect copyright.
 - Avoid creating harmful or offensive images.
 - Never misuse AI images.
 - Give credit when required.
 - Use AI for educational and creative purposes.
 - Respect privacy and avoid creating misleading images.
-

Best Practices for Image Prompting

Students should:

- Be clear and specific.
 - Mention the subject.
 - Add background details.
 - Specify colors and style.
 - Include lighting.
 - Mention image quality.
 - Review AI-generated images.
 - Improve prompts if necessary.
-

Real-Life Applications

Image prompting is useful for:

- School projects
 - Poster competitions
 - Presentation slides
 - Logo design
 - Graphic design
 - Digital marketing
 - Social media content
 - Storybook illustrations
 - Website graphics
 - Educational materials
-

Career Opportunities

Image prompting skills are valuable for careers such as:

- Graphic Designer
- AI Image Designer
- Digital Artist
- UI/UX Designer
- Brand Designer
- Social Media Designer
- Marketing Executive

- Content Creator
 - Animator
 - Creative Director
-

Revision

Students should revise:

- Image Prompt
 - Components of a Good Prompt
 - Poster Prompt
 - Logo Prompt
 - Illustration Prompt
 - Social Media Prompt
 - Prompt Improvement
 - Ethical AI Image Creation
-

Did You Know?

- AI can create posters, logos, illustrations, and digital artwork from simple text prompts.
 - Adding details such as colors, lighting, style, and background improves image quality.
 - Designers often refine image prompts several times to achieve the desired result.
 - AI image generation is widely used in education, advertising, marketing, publishing, and digital design.
-

Points to Remember

- An image prompt is a text description used to generate an AI image.
- Detailed prompts produce better-quality images.
- Include the subject, background, style, colors, lighting, mood, and quality.
- Improve prompts by adding more descriptive details.
- Use AI-generated images ethically and responsibly.